

Sleepiness Function Estimation Strategies

Structural Context

The sleepiness function $\phi_{mt}(\mathbf{s}_{mt})$ governs the fraction of depositors in market m at time t who do **not** re-optimize their deposit allocation, i.e., who remain "asleep" and mechanically roll over their balance. The structural equation (Eq-11/12 of Egan et al., 2025) posits:

$$D_{jmkt} = \phi_{mt}(\mathbf{s}_{mt}) \cdot \widetilde{D}_{jmkt-1} + (1 - \phi_{mt}(\mathbf{s}_{mt})) \cdot \hat{D}_{jmkt}$$

where $\widetilde{D}_{jmkt-1} = (1 + r_t^f - p_{jkt}) \cdot D_{jmkt-1}$ is the **no-rebalancing counterfactual** deposit (lagged deposits grown at the net return), and $\hat{D}_{jmkt}$ is the BLP-implied demand.

Re-arranging and substituting $\hat{D}_{jmkt}$ out via fixed effects:

$$D_{jmkt} = \phi(\mathbf{s}_{mt}) \cdot \widetilde{D}_{jmkt-1} + \alpha_{jmk} + \varepsilon_{jmkt}$$

All estimation scripts estimate this equation via **within-group (entity-demeaned) OLS** on:

Variable	Role
deposit_balance	Dependent variable D_{jmkt}
nr_lagged_dep $\equiv \widetilde{D}_{jmkt-1}$	Core regressor (no-rebalancing counterfactual)
State-space interactions $\mathbf{s}_v \cdot \widetilde{D}_{jmkt-1}$	Heterogeneity in ϕ
v_hat , v_hat_2 , v_hat_3	Control Function (CF) residuals (when IVs used)

The coefficient on nr_lagged_dep (the "constant" state variable) is the intercept of ϕ ; the coefficients on the interaction terms trace how ϕ varies with observables.

Common Econometric Infrastructure

All five scripts share the following building blocks.

1. Data Construction

- Load market_panel.csv ; reshape from wide to long if deposit types are stored as columns (dep_a1 , dep_a2 , ...).
- Drop deposit type 3** (demand deposits / current accounts — not subject to sleepiness friction).
- Construct entity ID: CodConglomeradoPrudencial $\times$ deposit_type $\times$ mca_code .
- Generate lagged variables: spread_qoq_lag , risk_free_qoq_lag , lagged_deposits .
- Build $\widetilde{D}_{jmkt-1}$: nr_lagged_dep $= (1 + r_{t-1}^f - p_{jk,t-1}) \times D_{jmk,t-1}$.
- Scale numerical magnitudes (deposits $\div 10^9$; GDP per capita $\div 10^4$; etc.) for numerical stability.

2. First Stage — Control Function Approach (CFA)

Endogeneity of the spread p_{jkt} is addressed via a **Petrin–Train (2010)** control function:

- Restrict** to endogenous deposit types ($k \in \{4, 5\}$ — CDBs and Letters of Credit).
- Regress** p_{jkt} on instruments $\mathbf{z}$ and exogenous state variables:

$$p_{jkt} = \mathbf{z}'_{jt} \boldsymbol{\pi} + \mathbf{x}'_{mt} \boldsymbol{\gamma} + v_{jkt}$$

- Extract** residuals $\hat{v}_{jkt}$ and their powers ($\hat{v}^2, \hat{v}^3$).
- These enter the second stage as **flexible CF corrections** — a polynomial in the first-stage residual.

3. Instrumental Variable Sets

Four IV strategies are defined across all scripts (identical IV blocks):

IV Label	Instruments
OLS	None — no CF correction
IV_CostShifters	personnel_cost_ratio_lag , admin_cost_ratio_lag , tax_cost_ratio_lag
IV_Wholesale	CostShifters + lci_lca_ratio_lag , wholesale_ratio_lag , indice_basileia_lag
IV_HausmanFull	Wholesale + leave_one_out_mean_spread (Hausman-type instrument)

4. Imbens & Kolesar (2016) Cluster Correction

All scripts apply a **Satterthwaite-type effective cluster count**, denoted G^* :

$$G^* = \frac{G}{1 + CV^2(N_g)}$$

where G is the nominal cluster count and $CV(N_g)$ is the coefficient of variation of cluster sizes. Highly imbalanced clusters (e.g., dominated by large banking conglomerates) inflate nominal G ; the correction ensures conservative inference by using $t(G^*)$ critical values.

5. Phi Aggregation

After estimation, $\hat{\phi}_{mt}$ is constructed for every observation by evaluating the estimated linear combination $\hat{\beta}' s_{mt}$. Two levels of aggregation follow:

- **Local** $\hat{\phi}_{mt}$: Mean across entities within market m and quarter t .
- **National** $\hat{\phi}_t$: Deposit-weighted average across all markets:

$$\hat{\phi}_t = \frac{\sum_m \hat{\phi}_{mt} \cdot M_{mt}}{\sum_m M_{mt}}$$

where M_{mt} is the market deposit stock.

Specification 1 — Baseline (B-Type Only, Linear, Post-2020 Break)

File: [estimation_1_sleep.py](file:///c:/Users/pedro/OneDrive/Documentos/Yale/Year%203%20(2024%20-%202025)/Open%20Finance/Open-Finance/Code/Egan_et_al_2025_Rep/estimation_1_sleep.py)

Purpose

Baseline sleepiness estimation using **B-type (branch-based) institutions only**. This is the primary specification for the Egan et al. (2025) replication.

Sample Restriction

- **B-type only:** Filtered by `CODMUN_IBGE != '0'` (entities with a geographic market code).
- Deposit types: $k \in \{1, 2, 4, 5\}$ (type 3 excluded).

State Variable Blocks

Three nested blocks, all including `post_2020` as a structural break dummy:

Block	States s_{mt}
Base	constant , post_2020
Macro	Base + gdp_per_capita , cadunico_families_per1000 , fraction_65plus , fraction_young , risk_free_qoq_lag
Tech	Macro + pix_users_pf_per1000 , connections_per100 , branches_per1000

Second Stage

Standard **within-group OLS** (entity fixed effects via demeaning). The regression is:

$$\tilde{D}_{j m k t}=\sum_{v \in \mathbf{s}} \beta_v \cdot\left(s_v \times \tilde{D}_{j m k t-1}\right)^{\sim}+\text { CF terms }^{\sim}+\tilde{\varepsilon}_{j m k t}$$

where $\tilde{}$ denotes entity-demeaned values.

Specification Grid

The script runs 4 IV sets $\times$ 3 state blocks = 12 specifications in parallel via `ProcessPoolExecutor` .

Outputs

- TeX tables for each specification (second stage + first stage).
- `cluster_diagnostics.json` : G , G^* , CV, top-5 clusters.
- `estimation_results.pkl` : Serialised `statsmodels` results.
- `market_panel_phi.csv` and `national_phi_t.csv` : Phi estimates.

Key Design Choices

- `post_2020` enters the state space, allowing the **level** of ϕ to shift after 2020 (PIX launch, COVID effects).
- Missing state variables imputed with column medians.
- Parallel execution across specifications for throughput.

Specification 3 — No Structural Break (B-Type Local + D-Type National)

File: [estimation_3_sleep.py](file:///c:/Users/pedro/OneDrive/Documentos/Yale/Year%203%20(2024%20-%202025)/Open%20Finance/Open-Finance/Code/Egan_et_al_2025_Rep/estimation_3_sleep.py)

Purpose

Robustness check that **omits `post_2020`** from the state space. Tests whether the post-2020 structural break is driving identified sleepiness patterns or is absorbed by other time-varying covariates.

Key Differences from Spec 1

1. **No `post_2020`** : The base block is just `['constant']` ; a secondary base block `Base_Selic` adds `risk_free_qoq_lag` .
2. **Dual estimation**: Runs **local** (B-type) and **national** (D-type) separately.
3. **Three national model options** for D-type firms:

National Option	Approach
Option 1	Standard within-group OLS (identical to local, but using D-type sample)
Option 2	Adds <code>log_total_assets_lag</code> cross-interactions: each state variable s_v gets interacted with both $\tilde{D}$ and $s_v \times \log_assets \times \tilde{D}$
Option 3	Reduces multi-dimensional state vector to a single PCA index via <code>sklearn.decomposition.PCA(n_components=1)</code> . The regression then uses only <code>nr_lagged_dep</code> and <code>interaction_pca_index</code>

State Variable Blocks

Block	States s_{mt}
Base	constant
Base_Selic	constant , risk_free_qoq_lag

Block	States s_{mt}
Macro	Base_Selic + gdp_per_capita , cadunico_families_per1000 , fraction_65plus , fraction_young
Tech	Macro + pix_users_pf_per1000 , connections_per100 , branches_per1000

National Data Construction

- D-type firms identified by `Source == 'IFDATA'` (national-level supervisory data).
- Deduplicated at `CodConglomeradoPrudencial × year × quarter`.
- Entity ID: `CodConglomeradoPrudencial × deposit_type` (no market code since D-type is national).

Plotting

Generates comparison plots overlaying local $\hat{\phi}^{Loc}$ and three national variants ($\hat{\phi}^{Nat,1}$, $\hat{\phi}^{Nat,2}$, $\hat{\phi}^{Nat,3}$) with 95% CI bands.

Why This Matters

- Removing the break dummy tests identification **purely through time-varying macro/tech covariates**.
- PCA index (Option 3) tests whether the identified variation is dominated by a single latent factor — if so, the multi-covariate specification may be over-parameterised.
- Asset-size interactions (Option 2) test whether sleepiness differs systematically between large and small D-type institutions.

Specification 4 — Pooled B + D (Linear, PIX Dummy)

File: [estimation_4_sleep.py](file:///c:/Users/pedro/OneDrive/Documents/Yale/Year%203%20(2024%20-%202025)/Open%20Finance/Open-Finance/Code/Egan_et_al_2025_Rep/estimation_4_sleep.py)

Purpose

Robustness check that **pools B-type and D-type firms** in a single regression. Uses `dummy_D_type` and `pix_exists` as structural controls instead of the `post_2020` break.

Key Differences from Spec 1

- Pooled sample:** Both B-type (`CODMUN_IBGE != '0'`) and D-type (`CODMUN_IBGE == '0'`) firms.
- D-type dummy** (`dummy_D_type`): Allows a level shift in ϕ for digital-only/national institutions.
- PIX existence dummy** (`pix_exists`): Equals 1 for $t \geq 2020Q4$, capturing the PIX instant payment system launch. More precisely timed than `post_2020`.
- D-type interaction terms:** `dummy_D_type × risk_free_qoq_lag`, `dummy_D_type × fraction_65plus`, `dummy_D_type × fraction_young` — allows differential sensitivity of B vs. D firms to monetary policy and demographics.
- Population-weighted state imputation for D-firms:** D-type firms lack municipality-level covariates. For each quarter, their state variables are set to the **population-weighted national average** of B-type firms' covariates.

State Variable Blocks

Block	States s_{mt}
Base	constant , <code>dummy_D_type</code> , <code>pix_exists</code>
Base_Selic	Base + <code>risk_free_qoq_lag</code> , <code>dummy_D_type × risk_free_qoq_lag</code>
Macro	Base_Selic + <code>gdp_per_capita</code> , <code>cadunico_families_per1000</code> , <code>fraction_65plus</code> , <code>fraction_young</code> , <code>dummy_D_type × fraction_65plus</code> , <code>dummy_D_type × fraction_young</code>
Tech	Macro + <code>pix_users_pf_per1000</code> , <code>connections_per100</code> , <code>branches_per1000</code>

Second Stage

Identical within-group OLS as Spec 1, but on the pooled sample.

Plotting

Produces time-series plots of $\hat{\phi}_t$ separately for B-type and D-type sub-populations. Weighted-variance confidence bands are computed using effective sample sizes ($n_{\text{eff}} = V_1^2/V_2$ where $V_k = \sum w_i^k$).

Why This Matters

- Tests whether the B-type and D-type sleepiness functions are **structurally similar** once controlled for size and demographic differences.
- The PIX dummy is a more precise event-study indicator than `post_2020`.
- Pooling increases statistical power, especially for identifying differential effects through interaction terms.

Specification 5 — Pooled B + D (Logistic Link, NLLS)

File: [estimation_5_sleep.py](file:///c:/Users/pedro/OneDrive/Documentos/Yale/Year%203%20(2024%20-%202025)/Open%20Finance/Open-Finance/Code/Egan_et_al_2025_Rep/estimation_5_sleep.py)

Purpose

Identical sample and state variables to Spec 4, but replaces the **linear second stage** with a **logistic link function** to enforce $\phi \in [0, 1]$.

Structural Modification

Instead of:

$$D_{jmkt} = \left(\sum_v \beta_v \cdot s_v \right) \tilde{D}_{jmkt-1} + \alpha_{jmk} + \varepsilon_{jmkt}$$

this specification uses:

$$D_{jmkt} = \Lambda \left(\sum_v \beta_v \cdot s_v \right) \tilde{D}_{jmkt-1} + \alpha_{jmk} + \varepsilon_{jmkt}$$

where $\Lambda(x) = \frac{e^x}{1+e^x}$ is the **logistic sigmoid**. This guarantees $0 \leq \hat{\phi}_{mt} \leq 1$ regardless of the estimated parameters.

Estimation Method

Non-Linear Least Squares (NLLS) via `scipy.optimize.least_squares`:

1. The objective function `nlls_objective` computes entity-demeaned residuals:
 - For each observation, compute $\hat{Y} = \Lambda(\mathbf{x}_i' \boldsymbol{\theta}) \cdot Z_i + \mathbf{CF}_i' \boldsymbol{\gamma}$.
 - Entity-demean $\hat{Y}$ via `np.bincount(entity_idx, weights=Y_hat)`.
 - Return $\tilde{y} - \tilde{\hat{Y}}$.
2. **Solver:** Trust Region Reflective (`method='trf'`) with **Cauchy loss** (`loss='cauchy'`) for outlier robustness.
3. **Max function evaluations:** 150 (deliberately capped to avoid overfitting in high-dimensional specifications).

Inference: Average Marginal Effects (AME)

Since the logistic link is nonlinear, raw coefficients $\hat{\boldsymbol{\theta}}$ have no direct marginal interpretation. The script computes **Average Marginal Effects** via the Delta Method:

- For **continuous** variables: $\text{AME}_k = \frac{1}{N} \sum_{i=1}^N \Lambda'(\mathbf{x}_i' \hat{\boldsymbol{\theta}}) \cdot \hat{\theta}_k$
- For **dummy** variables: $\text{AME}_k = \frac{1}{N} \sum_{i=1}^N \left[\Lambda(\mathbf{x}_i^{(k=1)} \hat{\boldsymbol{\theta}}) - \Lambda(\mathbf{x}_i^{(k=0)} \hat{\boldsymbol{\theta}}) \right]$

Standard errors of AMEs are obtained via **numerical Jacobian** of the AME function w.r.t. $\hat{\boldsymbol{\theta}}$, then $\text{Var}(\text{AME}) = J \cdot \hat{\Sigma} \cdot J'$.

Phi Construction

Uses the **native (un-transformed) parameters** `params_native` to evaluate the sigmoid:

$$\hat{\phi}_{mt} = \frac{1}{1 + \exp\left(-\hat{\boldsymbol{\theta}}' \mathbf{s}_{mt}\right)}$$

This ensures $\hat{\phi} \in [0, 1]$ by construction.

NonLinearResults Container

A custom results class stores:

- `params` : AME values (for comparability with linear specs).
- `params_native` : Raw logistic coefficients $\hat{\boldsymbol{\theta}}$.
- `bse` , `tvalues` , `pvalues` : Delta-method SEs and IK2016-corrected inference.
- `G_star` : Effective cluster count.

Why This Matters

- The linear specification can produce $\hat{\phi} < 0$ or $\hat{\phi} > 1$ — economically nonsensical. The logistic link eliminates this.
- Cauchy loss provides **robustness to outlier markets** (e.g., very small or very large banks dominating a municipality).
- AME reporting allows direct comparison of effect magnitudes with the linear specifications.

Specification 6 — Pooled B + D + Institutional Heterogeneity (Linear + Logistic)

File: [estimation_6_sleep.py](file:///c:/Users/pedro/OneDrive/Documents/Yale/Year%203%20(2024%20-%202025)/Open%20Finance/Open-Finance/Code/Egan_et_al_2025_Rep/estimation_6_sleep.py)

Purpose

Extends the pooled estimation by introducing **institutional type heterogeneity**: cooperative banks (`is_coop`) and state-owned banks (`is_state_owned`). Runs **both** linear (from Spec 4) and logistic (from Spec 5) second stages for each specification.

Key Additions

1. **Ownership dummies:** `is_coop` , `is_state_owned` enter the state space directly.
2. **Two alternative interaction structures** (CLI-selectable via `--alt`):

Alternative 1: Full Interactions

Every state variable s_v (except `constant`) gets interacted with both ownership dummies:

$$\phi_{mt} = \beta_0 + \beta_{\text{coop}} \cdot \mathbf{1}_{\text{coop}} + \beta_{\text{state}} \cdot \mathbf{1}_{\text{state}} + \sum_v [\beta_v + \beta_{v,\text{coop}} \cdot \mathbf{1}_{\text{coop}} + \beta_{v,\text{state}} \cdot \mathbf{1}_{\text{state}}] s_v$$

This is a **fully saturated** model w.r.t. institutional type $\times$ state variable interactions. For the Tech block, this can produce $\sim 30+$ parameters.

Alternative 2: Targeted Interactions (Default)

Only demographics and monetary policy interact with ownership:

- `is_coop` $\times$ `risk_free_qoq_lag` , `is_state_owned` $\times$ `risk_free_qoq_lag`
- `is_coop` $\times$ `fraction_65plus` , `is_state_owned` $\times$ `fraction_65plus`
- `is_coop` $\times$ `fraction_young` , `is_state_owned` $\times$ `fraction_young`

This is **more parsimonious** and economically motivated: cooperative banks serve older/rural demographics; state-owned banks may have different interest-rate pass-through.

Dual Second Stage

Each specification is run with **both** the linear within-group OLS (identical to Spec 4) and the logistic NLLS (identical to Spec 5). This produces paired estimates for comparison:

Model Type	Second Stage	ϕ Bounds	Inference
linear	Within-group OLS	Unbounded	Cluster-robust SEs + IK2016
logistic	NLLS with sigmoid	[0, 1]	AME + Delta-method SEs + IK2016

Parallelisation

Uses `joblib.Parallel(n_jobs=4)` (rather than `ProcessPoolExecutor` used elsewhere) to manage memory under the expanded parameter space.

Phi Construction

Identical to Specs 4 and 5 respectively, depending on `model_type` :

- **Linear**: $\hat{\phi}_{mt} = \hat{\beta}' s_{mt}$
- **Logistic**: $\hat{\phi}_{mt} = \Lambda(\hat{\theta}' s_{mt})$

Why This Matters

- Brazil's banking system has substantial heterogeneity: cooperatives (e.g., Sicoob, Sicredi) serve primarily rural markets with older depositors; state-owned banks (BB, CEF) have implicit government guarantees.
- If cooperatives exhibit **higher** sleepiness (less withdrawal risk), this has direct implications for deposit pricing and market power.
- Running both linear and logistic provides a **specification sensitivity check** on the functional form assumption.

Summary Comparison

Feature	Spec 1	Spec 3	Spec 4	Spec 5	
Sample	B-type only	B (local) + D (national)	Pooled B+D	Pooled B+D	
Post-2020 break	✓ (<code>post_2020</code>)	X	X (<code>pix_exists</code> instead)	X (<code>pix_exists</code> instead)	X (<code>p</code>
D-type dummy	X	N/A (separate)	✓	✓	
Institutional heterogeneity	X	X	X	X	i
Second stage link	Linear	Linear	Linear	Logistic	
National model options	—	3 variants	—	—	
PCA dimension reduction	X	✓ (Option 3)	X	X	
Asset-size cross-terms	X	✓ (Option 2)	X	X	
AME reporting	X	X	X	✓	
Parallelisation	ProcessPoolExecutor	ProcessPoolExecutor	ProcessPoolExecutor	joblib	

Feature	Spec 1	Spec 3	Spec 4	Spec 5	
Output directory	rout_1/	rout_3/	rout_4/	rout_5/	
State blocks	Base/Macro/Tech	Base/Base_Selic/Macro/Tech	Base/Base_Selic/Macro/Tech	Base/Base_Selic/Macro/Tech	Base/B

 Note

There is no `estimation_2_sleep.py` in this repository. Specification numbering skips from 1 to 3.

References

- Egan, M., Hortaçsu, A., & Matvos, G. (2025). *Deposit Competition and Financial Fragility*. Working Paper.
- Petrin, A. & Train, K. (2010). A Control Function Approach to Endogeneity in Consumer Choice Models. *Journal of Marketing Research*, 47(1), 3–13.
- Imbens, G. W. & Kolesar, M. (2016). Robust Standard Errors in Small Samples: Some Practical Advice. *Review of Economics and Statistics*, 98(4), 701–712.